

How to Write Better CHI Papers (with LaTeX in Overleaf)

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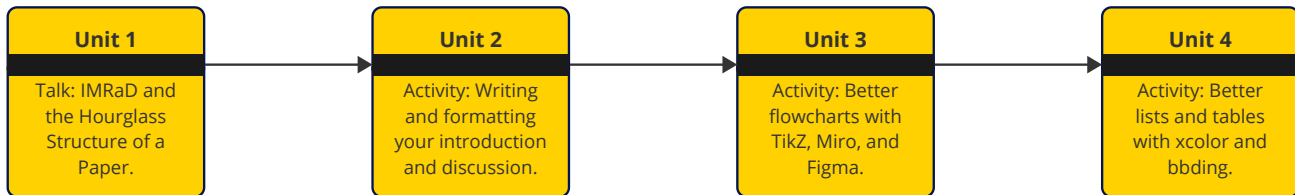


Figure 1: Overview of the four course units proposed for this online course. The audience will first learn about writing a CHI paper in general and then get specific tips and tricks for writing papers in Overleaf.

ABSTRACT

Writing and organizing research papers is a valuable skill that can make or break your academic career. This course will help you improve your skills in writing research papers for publication at CHI. In the past five years, my writing course at CHI has introduced you to everything you wanted to know about writing papers. Now, it is time to take your skills even further. This course is now also about how to use LaTeX and Overleaf to format your papers so that they are easy to read and have the most practical impact. It is broken up into four 75-minute online units that will help you structure your paper’s research content and typographic design. The goal of the course is to help you write something that makes a contribution to the field of human-computer interaction and can be understood by other HCI researchers through engaging formatting and concise language.

CCS CONCEPTS

• **Applied computing** → *E-learning*; • **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Writing, Submission Process, Research Methods, LaTeX, Clarity

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1 BENEFITS

The main benefit of the course is that participants will enhance their credibility and reputation as a researcher by producing high-quality papers. Students will learn practical writing skills that enable them to “trim the fat” in their writing and focus on bringing essential information about their research across in the first course unit (some of the advice being taken from essential style guides [2, 13, 15, 19]). They will also learn how a CHI paper fits into the traditional IMRAD (Introduction, Methods, Research, And Discussion) model. Some pointers for carving out a research niche using the CaRS model from Swales [14] will be given to help participants think about how to write an introduction. We will also briefly touch upon the contribution types for the CHI conference as outlined by Wobbrock and Kientz [18]. The other units (see Figure 1) are more focused on practical tips of paper writing in Overleaf.

Specifically, I aim for participants to take away the following skills from this course:

- ✓ Understanding the IMRAD (see Figure 2) structure of a paper and how to highlight contributions throughout the paper.
- ✓ Working in LaTeX/Overleaf to create professional, consistently formatted manuscripts.
- ✓ Guiding the reader through formatting of essential sections and crucial text in the paper.
- ✓ Creating more structured diagrams and better flowcharts of their research processes and systems.
- ✓ Improving typography and visuals for better readability and impact.
- ✓ Developing the ability to structure and present complex information in a clear and logical manner.
- ✓ Presenting information chunks in tables and lists more clearly.
- ✓ Improving the overall quality and clarity of your writing, which can increase the chances of your paper being accepted for publication.

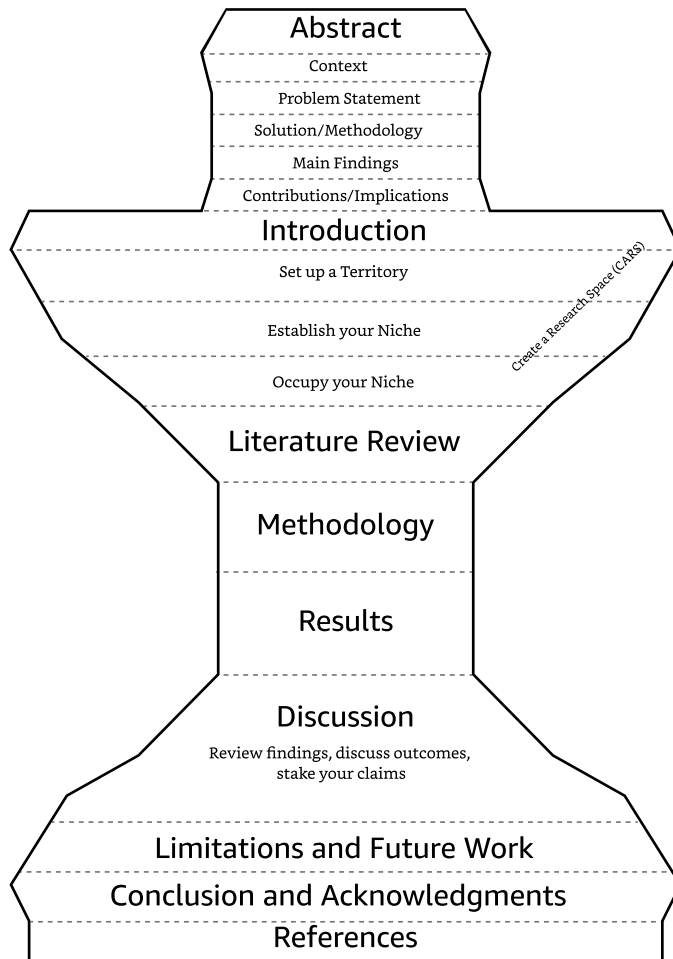


Figure 2: The research writing hourglass structure (including the CARS - Create a Research Space model) mapped onto the sections of a CHI paper. (From: <https://chicourse.com>)

This course is an excellent opportunity for CHI attendees to learn more about the process of writing CHI papers and gain an understanding of how to use $\LaTeX$ when working in Overleaf. People who attend will not only learn more about this type of scientific writing in general, but they will also learn specific tips and tricks for formatting documents in $\LaTeX$ within Overleaf. With this new knowledge, authors can be sure that their papers follow all the rules needed to be published at the CHI conference [12].

2 INTENDED AUDIENCE

This course introduces the structure of writing a research paper. Then, it includes tips about how to typeset the paper using $\LaTeX$ in the editing software Overleaf. A core understanding of $\LaTeX$ is beneficial. Researchers that would like to improve presenting their work with Overleaf are a great fit. Overall the following people will be a great audience for the course:

- Junior researchers seeking additional information about publishing at CHI.

- All academics who are interested in learning a few shortcuts and formatting techniques for $\LaTeX$ using Overleaf for vital sections of their papers.
- Experienced researchers who want to improve the presentation and formatting of their papers for publication at CHI or other conferences.
- Graduate students who are preparing to submit their first research paper and want to ensure that it is well-written and well-formatted.
- Professors and instructors who want to learn more about using $\LaTeX$ and Overleaf to create professional-quality documents for their own research or for teaching purposes.
- Technical writers and other professionals who want to learn more about using $\LaTeX$ and Overleaf for creating documents with complex formatting requirements.
- Anyone interested in improving their skills in research writing and formatting for publication in academic or professional settings.
- Anybody wishing to publish at the CHI conference.

3 PREREQUISITES

There are no prerequisites for this course other than visiting the online course materials¹ before the conference to familiarize oneself with some of the course concepts and to listen to the interviews². The participants are expected to have a computer with a stable Internet connection, online video software with microphone and webcam available. It is also recommended that they have accounts and logins created for Overleaf, Notion, and Miro before the beginning of the course.

4 CONTENT

The course is designed to provide a comprehensive guide on how to write better CHI papers. The four online units (see Table 1) will take you through all the necessary steps for writing an effective paper. In the first unit, you will learn about IMRaD and CHI paper structures, gain knowledge of research methods [5], and develop your argument effectively. In the second unit, we will focus on how to write and format the most important parts of your paper: the introduction and discussion. You will learn techniques such as rhetorical structures that will help make your work stand out from other submissions. In Unit 3, we talk about how to make flowcharts with TikZ, Miro, and Figma. This section introduces tools and frameworks like these that can be used to show data visually in a way that is interesting. Finally, in unit four, we dive into building better lists and tables with $\LaTeX$. Here, you will learn how to use this markup language (and online tools) to present information in a document in an efficient and attractive way. By completing all four modules of this course successfully, you should have acquired enough knowledge to produce high-quality CHI papers that stand out from others. At the start of the first unit, the participants are introduced to the course instructor and the course goals:

¹<http://chi2023.chicourse.com>

²<https://anchor.fm/how-to-write-chi-papers>

- ✓ Learn the structure of CHI papers to be able to systematize your writing
- ✓ Understand how to communicate important information rapidly using $\LaTeX$ and Overleaf
- ✓ Create better visuals and tables to communicate results and takeaways more effectively

Table 1: Structure of four online course units.

Unit 1 CHI Paper Structure	
25 mins	Introduction, Goals, and Icebreaker activity
50 mins	Lecture and discussion on IMRaD hourglass structure and CHI contributions
Unit 2 Writing and Formatting Intro and Discussion	
15 mins	Review and setup of Overleaf for everyone
60 mins	Guided dialogue and seminar-style discussion about formatting intro and discussion section
Unit 3 Better Diagrams and Flowcharts	
25 mins	Overview and guided tutorial of TikZ basics with templates
25 mins	Overview and guided tutorial of Miro
25 mins	Overview and guided tutorial of Figma
Unit 4 Better Lists and Tables	
10 mins	Introduction of xcolor and examples
10 mins	Introduction of bbding and examples
20 mins	Formatting beautiful tables with examples
20 mins	Practice session with tables and lists
15 mins	Debrief, discussion, and feedback

4.1 Unit 1: CHI Paper Structure

This lecture unit will show people how to structure their research papers for the CHI conference in a very clear way. It will talk about things like how to break down a research problem into a format that can become the skeleton of a manuscript and how to position your research [11] so that it has the best chance of being accepted by reviewers [4]. Additionally, it will provide an overview of what reviewers are looking for in terms of content, format, and presentation when assessing CHI papers [16, 17]. I will show video interviews I did with senior CHI researchers and talk about what I learned from my podcast. I focus primarily on introducing the hourglass paper structure and the types of contributions that are suitable for these types of publications. But I also tell them what questions they need to ask to write a well-organized and interesting paper [1]. These questions help students find good research questions and point out where to seek evidence to back up their arguments. They can also use this information to develop an effective outline for their paper that goes into detail about all aspects of the topic. Asking key questions is essential when creating strong CHI papers because it helps ensure that all necessary components have been addressed clearly and concisely. If participants follow these structuring rules, their paper should have a better chance of being accepted and published at the CHI Conference. Specifically, we will go over five research questions that every scientific paper must answer:

- 👉 **Context:** Why did you do it?
- 👉 **Problem:** What did you do?
- 👉 **Solution:** How did you do it?

👉 **Results:** What did you find out?

👉 **Implications:** What does it all mean?

4.2 Unit 2: Writing and Formatting Intro and Discussion

At the start of this unit, we will ensure that everyone has their Overleaf account setup and is prepared to join in. Writing the introduction and discussion sections of a CHI paper can be challenging. We will discuss helpful tips for formatting these essential paper sections together. In the activity, I will explain how to come up with research questions that are both specific enough to cover the topic of your paper and broad enough to interest academic readers. You will also practice how to make contribution lists that show how valuable your work is and how to give a short summary of its results. Furthermore, it is essential for authors to know how to best structure their takeaways so that readers understand the implications of their findings quickly and easily. We will practice this. Finally, we will go over effective heading structures for organizing writing within these two sections to make sure all information is presented logically and clearly communicates its purpose. As a reference point, examples from past papers will be supplied [3].

4.3 Unit 3: Better Diagrams and Flowcharts

Diagrams and flowcharts that show the results and other research process information well are crucial in a paper to help explain the research that was done. In this unit, we will look at some examples of successful diagrams and flowcharts used in CHI papers. We will then go over how to format two key example types: a research study sequence diagram and a Prisma flowchart using three popular programs: TikZ, Miro, and Figma. Each program has its own benefits when creating visuals for your work, so you can choose whichever works best for your project's needs. If you use these tools correctly, you can make strong visuals that will help your paper be more clear and organized.

4.4 Unit 4: Better Lists and Tables

In LaTeX, using colour formatting and symbol fonts can help your work stand out from the crowd. Colour formatting allows you to highlight key points, while symbol fonts can be used to create a unique look for lists or tables. In this unit, we will talk about how to use both of these techniques to make your CHI paper easier to read and visually more structured. We will practice with some tips for making sure your work looks professional and polished before submitting it for publication. We will focus primarily on packages approved for The ACM Publishing System (TAPS), but we will also discuss options beyond TAPS that can really make table formatting unique (such as booktabs). At the end of this unit, there will be a practice session where everyone can show off what they have learned. This will provide an opportunity for us to review and make sure that each student has developed a strong understanding of how to write and format better CHI papers. Afterward, there will be time for all participants to give feedback about their experience in the course so far. Taking this time to reflect on what has been done helps you and me ensure everyone gets the most out of the course and leaves with increased confidence in their ability to produce high-quality written work.

5 PRACTICAL WORK

The four 75-minute units of the course will cover different parts of writing papers. As mentioned, the first unit will start with an online lecture about writing papers. This will be followed by interactive exercises in Overleaf for practical follow-along editing. Participants are encouraged to share their own Overleaf documents during class so that everyone can learn from each other and provide feedback on one another's work. Online exercises with post-it notes (using Miro boards) will be used to review approaches and make learning easier. The goal of the coursework is to find a balance between participants' discussions and their practical writing in Overleaf so that they can enhance both their theoretical understanding and practical skill set when it comes to crafting effective CHI papers.

6 INSTRUCTOR BACKGROUND

Lennart E. Nacke, Ph.D., is a tenured full Professor for Human-Computer Interaction and Game Design at the University of Waterloo. He has taught the previous five iterations of this course at CHI since 2017 Nacke [6, 7, 8, 9, 10] and provides a version of this writing course on his website for purchase: <https://chicourse.com>. He runs the successful *How to Write CHI Papers* podcast and an interview series about writing CHI papers on YouTube. He has served on the SIGCHI program and steering committees and taught University graduate classes on HCI research methods. Dr. Nacke has co-organized many workshops for CHI over the past five years; he also chaired the CHI PLAY 2014 and Gamification 2013 conferences, served as the technical program co-chair for CHI PLAY 2015 and CHI Games and Play subcommittee co-chair for CHI 2017, INTER-ACT 2019 and CHI PLAY 2022 Full Papers Co-Chair and was the chair of the CHI PLAY steering committee until 2018. He has also reviewed hundreds of papers and had lots of his own submissions rejected from CHI.

7 RESOURCES

Participants can use the following resources to learn about the principles of writing and publishing successful CHI papers:

- ✚ How to Write CHI Papers Website: <https://chicourse.com>
- ✚ How to Write CHI Papers Podcast
- ✚ How to Write CHI Papers Mailing List
- ✚ How to Write CHI Papers Interview Series
- ✚ How to Write Better Research Papers Online Course

8 ACCESSIBILITY

All materials will be available as recordings and written transcripts so that everyone can take part. This will enable those with hearing impairments to access the content without difficulty. The instructors will also take extra steps to make sure that website materials are accessible for all users by providing meaningful ALT text for images and using colourblind-safe schemes. The subtitles will make it easy for everyone to follow along with online videos that were made during lessons. These measures taken together help guarantee a more inclusive course experience.

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